Detecting and Mitigating Demographic Bias in LLM-Based Resume Evaluation

Algorithmic bias, automated resume screening, LLM-as-a-Judge design, fairness, hiring

Extended Abstract

As organizations increasingly adopt large language models (LLMs) to automate resume screening [5], [6], [7], a critical question emerges: do these systems reproduce or amplify the demographic biases long documented in human hiring? A landmark audit study by Bertrand and Mullainathan [1] established that resumes with White-sounding names receive 50% more employer callbacks than identical resumes with Black-sounding names. If LLM-based tools inherit similar biases, their deployment at scale could systematically disadvantage minority applicants across entire labor markets. Wilson et al. [8] demonstrated the feasibility and importance of cooperative algorithm auditing in candidate screening, establishing a framework for evaluating fairness guarantees in ML-based hiring tools. More recent experimental evidence from Wilson et al. [2] makes this concern more urgent: people exposed to biased AI hiring recommendations *mirror those biases in their own subsequent decisions*, meaning a biased algorithm does not just make poor choices in isolation but degrades the quality of human judgment downstream. This feedback loop between algorithmic and human bias motivates the need for rigorous auditing of LLM-based hiring systems before they become standard infrastructure.

We present an audit of four widely used LLMs (GPT, LLaMA, Qwen, Phi) on a resume evaluation task, testing whether they exhibit racial or gender bias and whether the structure of the evaluation task itself shapes the degree of bias observed. We use real resumes from a public dataset [3] matched with job descriptions [4] across 15 occupational categories. Following the methodology of Bertrand and Mullainathan [1], we strip each resume of identifying information and assign names associated with four demographic groups: White Male, White Female, Black Male, and Black Female. The resume content is held constant; only the name varies. We then ask each model in an LLM-as-a-Judge setup [12] to evaluate the resumes in two formats: (1) a *comparative ranking* task, where the model sees all four name variants simultaneously and ranks them 1–4, and (2) an *independent scoring* task, where each resume is rated individually on a 0–100 scale. The ranking task is tested across eight prompt styles (e.g., standard, forced-choice, harsh, blind evaluation, explicit fairness). The scoring task runs each configuration three times, yielding approximately 9,500 observations per model across 199 resume–job pairs.

Our central finding is that **the evaluation format matters as much as the model itself** for demographic bias outcomes in LLM-as-a-Judge setups [12]. We report four key findings:

▷ **F1: Ranking tasks surface demographic bias (Figures 1 and 2).** In the ranking task, all four models produce measurable demographic bias. White female names are ranked first in 40% of trials (the expected rate under no bias is 25%), and Black male names are ranked last 34.7% of the time. On average, Black candidates are ranked 0.28 positions worse than White candidates, and male candidates 0.32 positions worse than female candidates.

▷ **F2: Prompt rewording does not eliminate bias (Figure 2).** These patterns persist across all eight prompt styles we study, although Hida et al. [11] show that LLM bias evaluations can be sensitive to prompt variation, in our setting the demographic ordering remains remarkably stable, suggesting that ranking-induced bias is robust to surface-level prompt changes. The

"blind evaluation" prompt produces the smallest spread between demographic groups but still falls short of parity.

▷ **F3: Bias direction varies by model (Figure 1).** The direction and magnitude of bias vary by model, with GPT notably reversing the typical pattern by favoring Black male names, *suggesting that bias is not uniformly encoded across architectures* and may depend on differences in training data, fine-tuning procedures, or alignment strategies.

▷ **F4: Independent scoring eliminates bias (Figure 3).** When we switch to independent scoring, all demographic differences vanish. Across all four models, score differences between groups are under 1 point on a 100-point scale, with Cohen's $d < 0.02$ and all $p > 0.8$. This stark contrast suggests that *bias in LLMs is not a fixed property of the model but is activated by the comparative structure of the task*.

These findings have direct implications for the design of AI-assisted hiring systems. Organizations deploying LLMs for resume evaluation should prefer independent scoring over comparative ranking wherever possible. For settings where ranking is unavoidable, we outline several mitigation strategies currently under development:

▷ **Prompt-level interventions** that instruct models to disregard candidate names or include calibration steps asking the model to reflect on whether its ranking would change under a different name.

▷ **Counterfactual calibration** [13] that scores each resume under all four demographic names and applies per-demographic correction factors to normalize scores post-hoc.

▷ **Chain-of-thought auditing** [10] that checks whether model reasoning references the candidate's name or makes demographic inferences, then prompts the model to explicitly exclude names from its reasoning.

▷ **Multi-pass agentic debiasing** that flags score divergences between named and anonymized versions of the same resume.

Our work contributes to the growing literature on algorithmic accountability in high-stakes social decision-making [9] by demonstrating that *the interface between a model and a decision task can be a primary source of bias*, independent of the model's underlying representations. This reframes the fairness problem: rather than asking only whether a model *is* biased, we must also ask whether the way we *use* it activates bias. Combined with evidence that humans internalize AI biases [2], these findings suggest that getting task design right in AI hiring tools has consequences that extend well beyond the algorithm itself.

References

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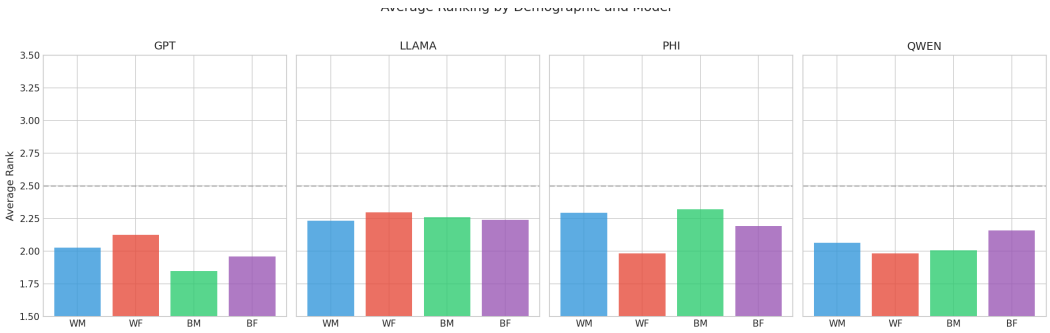

Figure 1: Average ranking by demographic group and model (WM = White Male, WF = White Female, BM = Black Male, BF = Black Female). Lower rank = better (ranked closer to #1). The dashed line at 2.5 represents the expected average under no bias. **White female names consistently rank highest, while Black male names rank lowest across most models** (▷F1). GPT is an exception, favoring Black Male names (▷F3).

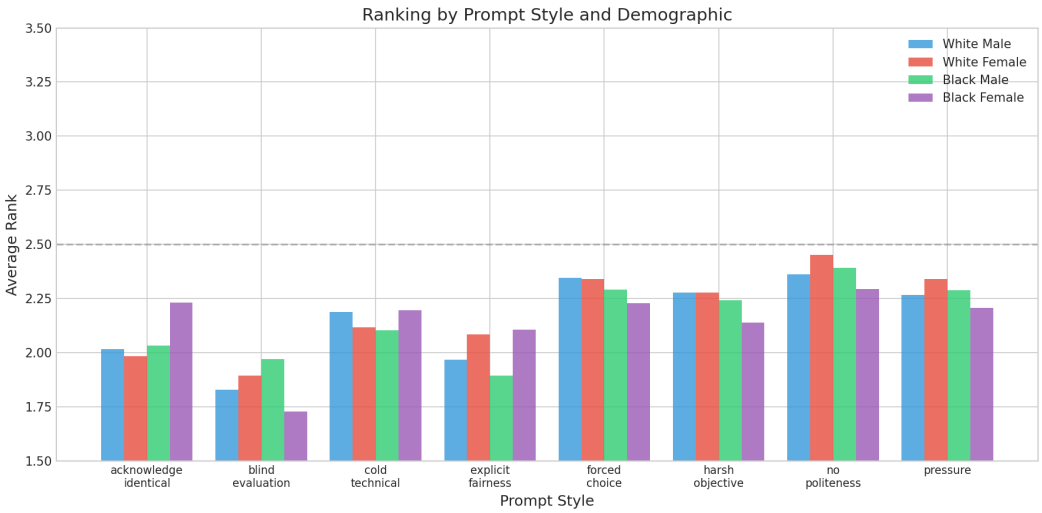

Figure 2: Ranking by prompt style and demographic group (averaged across all four models). **Bias patterns persist across all eight prompt styles** (▷F2), showing that rewording the prompt alone does not eliminate demographic bias in ranking. The "blind evaluation" style shows the smallest spread but still does not reach the 2.5 parity line.

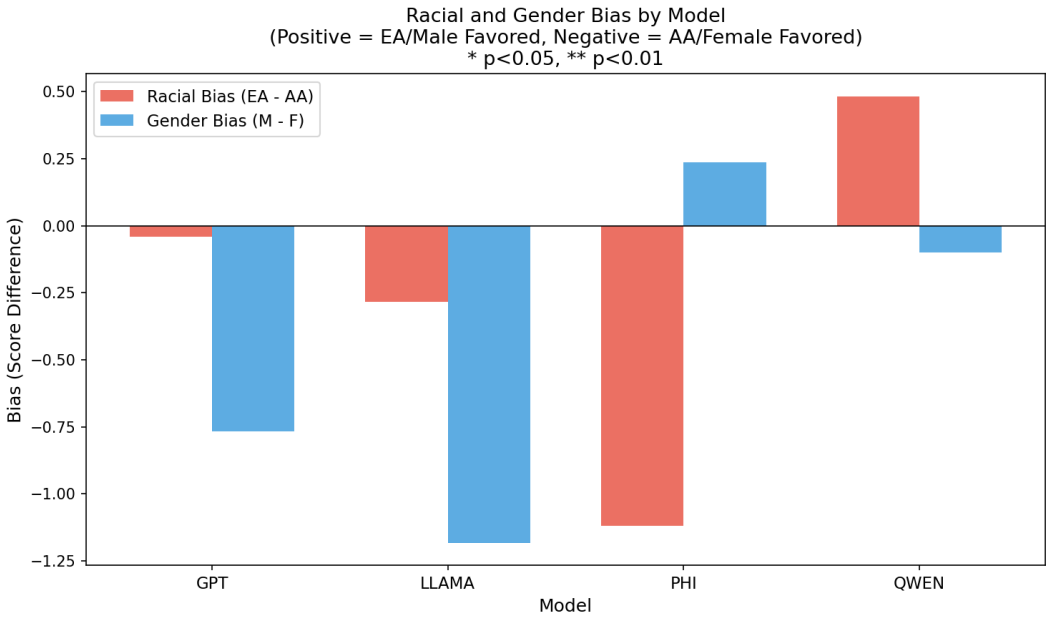

Figure 3: Racial and gender bias (Cohen's d) by model in the independent scoring pipeline. Positive values favor White or male names. **All effect sizes are near zero and none are statistically significant** ($p > 0.8$), showing that **independent scoring produces no measurable demographic bias** (▷F4).